

Smart Hospital - Project TODO

Phase 1: Database & Schema

- ✓ Design patient table with ID, name, email, medical history
- ✓ Design analysis_results table with patient_id, diagnosis, confidence_score, timestamp, doctor_name, analysis_type
- ✓ Design radiology_analysis table (brain, chest, bone) with image_url, model_output, confidence
- ✓ Design lung_cancer_analysis table with risk_factors, prediction, confidence
- ✓ Create and apply database migrations

Phase 2: Authentication & User Management

- ✓ Setup user authentication (already in template)
- ✓ Create doctor profile page with name and credentials
- ✓ Implement session persistence
- ✓ Add role-based access control (doctor, admin)

Phase 3: Radiology Sections

- ✓ Create Brain Tumor detection section
 - ✓ Image upload interface (DICOM, PNG, JPG)
 - ✓ Image preview before analysis
 - ✓ ML model integration for brain tumor detection
 - ✓ Display diagnosis with confidence score
- ✓ Create Chest Disease detection section
 - ✓ Image upload interface
 - ✓ Image preview
 - ✓ ML model integration
 - ✓ Display results with confidence
- ✓ Create Bone Fracture detection section

- ✓ Image upload interface
- ✓ Image preview
- ✓ ML model integration
- ✓ Display results with confidence

Phase 4: Lung Cancer Detection

- ✓ Create risk factor input form (smoking status, age, family history, etc.)
- ✓ Implement ML model for lung cancer prediction
- ✓ Display confidence-based prediction output
- ✓ Save results to database

Phase 5: Patient Records & History

- ✓ Create patient records database schema
- ✓ Build patient history page showing all past analyses
- ✓ Display all 5 required fields: patient_id, diagnosis, confidence_score, timestamp, doctor_name
- ✓ Add filtering and search functionality
- ✓ Add export/print functionality for medical records

Phase 6: Dashboard & Navigation

- ✓ Create elegant dashboard layout with sidebar
- ✓ Implement persistent sidebar navigation
- ✓ Link all radiology sections
- ✓ Add quick access buttons for common tasks
- ✓ Create professional visual design

Phase 7: Docker & Deployment

- ✓ Create Dockerfile for application
- ✓ Create docker-compose.yml for full stack (app + database)

- ☒ Add environment configuration
- ☒ Test local deployment
- ☒ Document deployment instructions

Phase 8: Testing & Refinement

- ☐ Test all radiology sections
- ☐ Test patient records saving and retrieval
- ☐ Test authentication flow
- ☐ Test Docker deployment
- ☐ Performance optimization
- ☐ UI/UX refinement for elegance and polish